THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** six	Two months each
2	**(b)** troposphere	Weather layer
3	**(a)** rise	Low pressure
4	**(c)** from sea to land	Land warmer
5	**(d)** sink	High pressure
6	**(a)** monsoon	Heavy rain
7	**(a)** greenhouse gases released by activities	Emission measure
8	**(c)** June to September	Rainy season
9	**(b)** mesosphere	Friction
10	**(c)** December and January	Winter

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Envelope of gases around the Earth (1); Held by gravity (1).	1+1
12	Greenhouse gases released by activity (1); Measured as total emissions (1).	1+1
13	Seasonal reversal of winds (1); Brings rain to India (1).	1+1
14	Water falling from clouds (1); Rain, snow, hail (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Up to 80 km (1); Meteors burn (1); Temperature falls with height (1).	1+1+1
16	Sea breeze (1); Land breeze (1); Both are daily (1).	1+1+1
17	Up to 50 km (1); Cloud-free (1); Ozone layer (1).	1+1+1
	OR: Average 12 km (1); Weather happens (1); Temperature falls with height (1).	1+1+1
18	Farming (1); Water supply (1); Economy and festivals (1).	1+1+1
	OR: Temperature and humidity (1); Precipitation (1); Pressure and wind (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Weather: today's rain (1); Climate: 30-year pattern (1); Weather changes fast (1); Climate stable (1); India has tropical monsoon climate (1).	1+1+1+1+1
	OR: Winter (1); Summer (1); Monsoon (1); Post-monsoon (1); Months and features (1).	1+1+1+1+1
20	Troposphere 12 km with weather (1); Stratosphere 50 km with ozone (1); Temperature falls then rises (1); Aeroplanes fly in stratosphere (1); Boundaries tropopause and stratopause (1).	1+1+1+1+1
	OR: Light near the poles (1); Solar wind (1); Borealis (1); Australis (1); Thermosphere (1).	1+1+1+1+1

Mistakes to avoid:

- **Weather** is the day-to-day state of the atmosphere; **climate** is the average weather over 30 years or more. A single rainy day is not climate.
- Sea breeze blows from sea to land in the **day** (land is warmer, low pressure on land); land breeze blows from land to sea at **night** and is weaker. Do not reverse them.
- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.
- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.

- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** atmosphere	Blanket of air
2	**(b)** carbon dioxide	Heat trapping
3	**(d)** human activities such as burning fossil fuels	Fossil fuels
4	**(c)** season	Arabic
5	**(c)** stratosphere	Protects from UV
6	**(a)** exosphere	Very thin
7	**(a)** precipitation	Falling moisture
8	**(b)** west	Named from direction
9	**(c)** nitrogen	78%
10	**(a)** 30 years or more	Long term

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Nitrogen (1); 78% (1).	1+1
12	Day-to-day state of the atmosphere (1); Changes quickly (1).	1+1
13	Stratosphere (1); Absorbs ultraviolet rays (1).	1+1
14	Moving air (1); From high to low pressure (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fossil fuels (1); Deforestation (1); Pollution (1).	1+1+1
16	Rain and drizzle (1); Snow and sleet (1); Hail (1).	1+1+1
17	Heavy rain (1); Weak embankments (1); Building near rivers (1).	1+1+1
	OR: Rises rapidly in temperature (1); Reflects radio waves (1); Auroras occur (1).	1+1+1
18	Weather short-term, climate long-term (1); Weather changes quickly (1); Climate average (1).	1+1+1
	OR: Troposphere, stratosphere (1); Mesosphere (1); Thermosphere, exosphere (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Greenhouse gases trap heat (1); Fossil fuels and deforestation (1); Effects on floods, droughts (1); Use renewable energy (1); Plant trees and save water (1).	1+1+1+1+1
	OR: Falls in troposphere (1); Rises in stratosphere (1); Falls in mesosphere (1); Rises in thermosphere (1); Exosphere very thin (1).	1+1+1+1+1
20	Troposphere (1); Stratosphere (1); Mesosphere (1); Thermosphere (1); Exosphere (1).	1+1+1+1+1
	OR: Agriculture (1); Water (1); Economy (1); Festivals (1); Floods or drought risk (1).	1+1+1+1+1

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- Order of layers from the surface: troposphere, stratosphere, mesosphere, thermosphere, exosphere. Ozone is in the **stratosphere**; weather and most water vapour in the troposphere.
- Temperature **falls** with height in the troposphere but **rises** in the stratosphere (ozone absorbs UV); wind blows from high pressure to low pressure.
- Nitrogen is about 78 per cent and oxygen about 21 per cent; do not swap them. Carbon dioxide is only about 0.04 per cent.

- South-west monsoon: June to September, most of India's rain; the north-east monsoon rains the Tamil Nadu coast. Greenhouse gases **warm** the Earth by trapping heat.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Atmosphere and Climate (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** south-west monsoon	Summer monsoon
2	**(a)** oxygen	21%
3	**(a)** a short time	Day-to-day
4	**(a)** summer	April-June
5	**(c)** saves electricity and cuts emissions	Reduces footprint
6	**(b)** troposphere	Clouds and rain
7	**(b)** absorb carbon dioxide	Carbon sinks
8	**(a)** high pressure to low pressure	Pressure difference
9	**(d)** Weather-Ready and Climate-Smart	Forecasting
10	**(d)** km per hour	Speed

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water vapour in the air (1); From evaporation (1).	1+1
12	Long-term change in weather pattern (1); Caused by humans (1).	1+1
13	Average weather over a long period (1); 30 years or more (1).	1+1
14	Troposphere (1); Contains water vapour (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	South-west monsoon (1); North-east monsoon (1); Different seasons (1).	1+1+1
16	Nitrogen 78% (1); Oxygen 21% (1); Argon 0.93% (1).	1+1+1
17	Floods and droughts (1); Melting glaciers (1); Rising sea level (1).	1+1+1
	OR: Walk or cycle (1); Save water (1); Reuse and recycle (1).	1+1+1
18	Crops damaged (1); Houses destroyed (1); People moved to relief camps (1).	1+1+1
	OR: Winter and summer (1); Monsoon (1); Post-monsoon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Day: sea breeze (1); Night: land breeze (1); Differential heating (1); Pressure difference (1); Moderates coastal climate (1).	1+1+1+1+1
	OR: Vasanta spring (1); Grishma summer (1); Varsha monsoon (1); Sharad and Hemanta (1); Shishira winter (1).	1+1+1+1+1
20	Nitrogen 78% (1); Oxygen 21% (1); Argon and CO2 (1); Water vapour (1); Dust (1).	1+1+1+1+1
	OR: Rivers Satluj, Beas, Ravi (1); Heavy monsoon rain (1); Weak embankments (1); Crop and house damage (1); Lessons for flood management (1).	1+1+1+1+1

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